Stopping the Data Flood: Post-Shannon Traffic Reduction in Digital-Twins Applications

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***Abstract*—Digital Twin (DT) implementations can reduce latency drastically in future communication systems for steering and control of cyber-physical systems (CPS). When the CPS and its DT periodically exchange information to repair possible desynchronisations between the two systems, this can lead to large data traffic flooding the networks. This work proposes a goal-oriented communication approach based on message identification (ID), which only sends repair data when necessary, but adds control traffic and a probability of not repairing some desynchronisations between the two systems. We compare optimal ID codes and a hash function for detection whether desynchronisation correction is necessary. Our method can reduce the traffic to within 0.3% of its optimal value while repairing 99.97% of all desynchronisations for 4 kbit of data and grows even better for larger data.**

***Index Terms*—digital twin, identification via channels, post-Shannon, goal-oriented communication**

I. Introduction

Many applications of future communication systems impose tight restrictions on the latency of communication networks. Applications such as the steering and control of Cyber-Physical Systems (CPS) or the haptic feedback for the tactile internet [1] require latencies below ten milliseconds. One proposed solution to deal with the latency constraints consists of deploying the so-called Mobile Edge Cloud (MEC) near the controlled entities and migrating the computing resources close to the entities, thus reducing the propagation delay of communication [2]. Furthermore, if the CPSs have to be controlled via long distances or if the controller uses the inputs of multiple CPSs, the network might employ the MEC as the underlying infrastructure to implement an in-network Digital Twin (DT) to reduce the latencies further [3]. A DT is a virtual model of the controlled entity and its environment used to represent the real system to decouple the control loops. The CPS's controller might make low latency decisions based on the DT, while the accuracy of the virtual model is updated with a less latency-constrained control loop with the physical system. For example, in tactile internet applications, the DT can be an interface between a human operator and a robot. The human controls the DT with low-latency haptic feedback and the DT is used to control the robot [3]. In the control loop between the CPSs and the DT in the MEC, we are interested in the communication's goal rather than the transmitted data. If our goal is to keep the DT in sync with the physical system's state, the former can periodically send its compressed state information to the latter. While current systems can be sufficiently dimensioned to accommodate for large traffic bursts, in the future the projected increase in connected devices and state space's entropy will eventually strain the finite available bandwidth. From the view of the post-Shannon communication approach of goal-oriented communication [4], it is only beneficial to transmit the state when the two systems are not in sync. Optimal traffic of the system can be achieved by only transmitting in those cases. Identifying whether the states match, or if a transmission is necessary, can be achieved using identification (ID) codes [5], which incur a traffic overhead and a probability of misidentification. We study the network traffic and reliability trade-off for an optimal ID code and a computationally cheap but suboptimal hash function. In our proposed approach, the physical system encodes its state using an ID code and communicates it with the DT. Only when the DT finds the states to be *desynced*, the DT transmits the full state information to *repair* the physical system's state. To the best of our knowledge, ID codes and hash functions have not been studied as a mechanism to reduce the traffic load of DT-assisted communication systems. The system's simulations show that for short block lengths the hash function and the optimal ID-code behave similarly in all measured metrics but computational complexity.

II. Problem Statement and Background

The investigated scenario is composed of two parties observing an environment and computing a state from their observations. Both parties have different input data and sometimes compute different states (desynchronise) with a certain probability. In practice, an autonomous robot could use different sensors to determine which objects are in its vicinity and compute its behaviour to achieve some objective. The other entity could be a DT in a MEC, which has access to traffic cameras, location information of other robots, and routes' traffic information and wants to provide the robot with what it thinks is the best next action. We consider the DT's information beneficial for the robot, but not to be mission-critical. Thus, it is acceptable if the price for a large reduction in traffic is a few desynchronisations not being repaired.

States are computed and communicated periodically. The robot and the DT do not always agree on the state since they use different input data and computational power. The state calculated by the DT and the one estimated locally by the robot often agree, but there is a probability p_{desync} of the two parties to compute different states, which depends on the scenario. Only in cases where the states do not match, there is merit in transmitting the DT's information to the robot to synchronise the states. More computing power allows the DT to use, for example, a more sophisticated algorithm to determine the best next action. Instead of always sending the DT state to the robot, we propose a method for transmitting less data using ID codes and transmitting the state only in cases where the robot does not determine the next best state on their own.

A. Message Identification for Traffic Reduction

Verifying whether the states match is an application of ID codes as introduced by Ahlswede and Dueck in [5]. Message ID is a post-Shannon communication task [4], [6], since the goal of the communication is not as by Shannon's definition to convey a message from one point to another [7], but for the receiver to decide in a binary hypothesis test whether the transmitter encoded the one state (*identity*) the receiver also computed. The test answers the question: do the robot and the DT match, or does the robot's state need to be *repaired*? The traffic reduction of our approach stems from this reformulation of the communication problem. In ID, the entities do not transmit enough bits to convey the full state; they transmit enough information to identify if the computed states match at a low margin of error.

When a receiver's design criterion is to reproduce a transmitted message, it is possible to transmit $\sim 2^{nC}$ different messages, where C is the channel capacity, and n is the block length, as Shannon showed in [7]. The design criterion of an ID receiver is different, as it is interested in identifying whether the sender transmitted a relevant message or not. In our scenario, this relevant message is the state computed by the MEC that is supposed to match the computed state at the robot. It is possible to identify $\sim 2^{2^{nC}}$ different ID messages using a randomised ID code [5], [6]. Using ID codes thus theoretically allows for exponential reductions in traffic at infinite block length. When operating at finite block lengths, the gains are reduced, and even in noiseless channels an error probability p_{err} of false positive identifications is introduced, i.e., a non-detected mismatch between the computed states. These so-called type 2 errors are unavoidable, but bound, for ID codes and consciously accepted to reduce traffic, as opposed to type 1 errors, which result from channel imperfections known from classical communication theory. In this paper, only error-free channels are considered and thus only type 2 errors will impair the performance. Since there can be no false negative IDs, the system never transmits repair data despite the states matching and thus avoids all unnecessary traffic. On the other hand, for false positive IDs, the system does not repair desynced robot's states. Type 2 errors thus reduce the total traffic more than desired, worsening the reliability.

B. Construction of ID codes

Tagging codes were proposed as the first explicit ID code construction in [8]. An ID message is encoded using a tagging code by randomly choosing a position from an error correction codeword, which is treated as a set of tags, each tag of the same size as the field size q of the error correction code. The position-tag pair is transmitted to the receiver, which encodes their own ID message into a tag using the received random position and the same ID code. The receiver conducts a hypothesis test by comparing its own tag with the received tag, verifying whether they match. The higher the distance d of the error correction codebook, the less likely it is to randomly pick a position where two different ID messages' codewords share the same tag value. Thus, for high distance codebooks the probability of two tags matching between different ID messages is limited to a maximum error probability of misidentification λ_2.

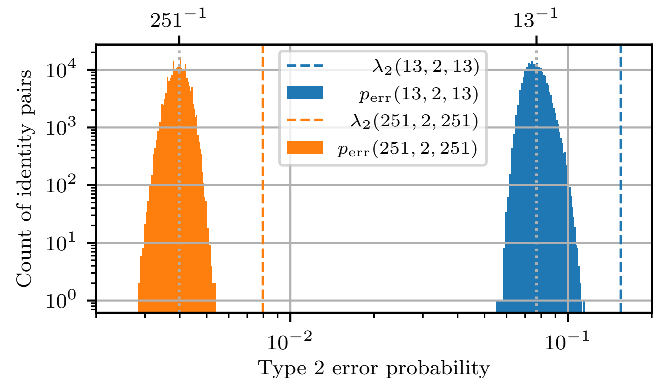

Fig. 1. Histogram of error probabilities of 200k ID pairs when using different RS-ID codes and their corresponding theoretical error bounds.

To create a high distance codebook, we use concatenated Reed-Solomon (RS) codes as they are maximum distance separable (MDS) codes. An $(n, k)_q$ RS code creates length n codewords from length k messages in base q. The minimal distance between codewords is $d = n - k + 1$. Note that we do not use RS as an error correction code as originally intended, but repurpose it for its distance properties, which are useful for ID code construction. An inner $(q-1, k_i)_q$ and an outer $(q^{k_i}-1, k_o)_{q^{k_i}}$ RS code are concatenated. The result is a $(q,\ k_i,\ k_o)$ RS code optimal for ID (RS-ID) [8], which can encode $q^{k_i k_o}$ IDs corresponding to a state size in bits of $n_{\text{data,max}} = k_i k_o \log_2(q)$. The authors of [9] extend RS-ID with optical orthogonal codes, outperforming the original construction in some regimes. To evaluate the traffic of transmitting one tag for verification, it is useful to use the size of the tag in bits: $n_{\text{ver}} = \lceil \log_2(q) \rceil$. The average type 2 error probability $\overline{p_{\text{err}}}$ is determined by how likely it is for a pair of IDs to create the same tag:

$$\overline{p_{\text{err,RS-ID}}} = \frac{1}{q} \approx 2^{-n_{\text{ver}}} \tag{1}$$

While $\overline{p_{\text{err}}}$ depends on the average overlap of tag sets of all pairs of IDs, due to the code construction p_{err} will never exceed λ_2 for any ID pair:

$$\lambda_2 = 1 - \frac{d_i d_o}{n_i n_o} = 1 - \frac{(q-k_i)(q^{k_i}-k_o)}{(q-1)(q^{k_i}-1)} \tag{2}$$

Fig. 1 shows as a histogram the simulated distribution of p_{err} for subsets of ID pairs using two RS-ID configurations. The error probability is distributed around a mean of $1/q$ as expected from equation (1). No ID pair's p_{err} exceeds λ_2 per the code construction, and none of the investigated pairs were found to even approach λ_2, since the pairs close to the bound are few, and an exhaustive search over all $\binom{q^{k_i k_o}}{2}$ ID pairs is prohibitive. For $q = 251$, the probabilities are estimates since it is prohibitive to sample more than a random fraction of all $(q-1)(q^{k_i}-1)$ possible positions in the tag sets.

For comparison with the optimal RS-ID code, we consider the hash function SHA-256 [10] as a suboptimal ID code. Like proper ID codes, hashes represent an input message as a fragment thereof, introducing potential confusions between which input resulted in the same output, which we call type 2 errors. We only use the first n_{ver} bits of the hashes digest as verifier, yielding the following average error probability:

$$\overline{p_{\mathrm{err,SHA-256}}} = 2^{-n_{\mathrm{ver}}} \tag{3}$$

In general, hash functions are bad ID codes and cannot achieve capacity, since they do not have a proven bound λ_2 on p_{err}. However, if the average error probability $\overline{p_{\mathrm{err}}}$ is a key performance indicator, bad ID codes can achieve high identification rates as described in [11], for SHA-256's arbitrary input size even arbitrary ID rates. At large block lengths, for which optimal ID codes are designed, hash functions are expected to be unfeasible due to high error rates.

TABLE I
RS-ID ENCODER CONFIGURATIONS AND THEIR THEORETICAL LIMITS

n_{ver}	n_{data}	q, k_o	k_i	$n_{\mathrm{data,max}}$	$\overline{p_{\mathrm{err}}}$	λ_2
4 bit	96 bit	13	2	96.2 bit	7.69%	14.88%
8 bit	96 bit	251	2	4001.7 bit	0.398%	0.795%
8 bit	4001 bit	251	2	4001.7 bit	0.398%	0.795%
12 bit	96 bit	4093	2	98.2 kbit	0.0244%	0.049%
12 bit	4001 bit	4093	2	98.2 kbit	0.0244%	0.049%
16 bit	96 bit	65521	2	210 Mbit	0.0015%	0.003%

III. Implementation

A. Model

We investigate the trade-offs between traffic reduction and the probability of the DT determining a false positive ID, resulting in a missed transmission of necessary state data to the robot, by simulating the scenario. The robot and its DT in the MEC can communicate with each other over an error-free channel at each simulation tick. In our scenario both parties are expected to derive the next state of the robot based on different input data. Thus, for each tick, the robot is assigned a new state uniformly-randomly selected from all possible states. The DT is set to the same state, or with the desync probability p_{desync} a different random state, which models the DT coming to a different conclusion than the robot. When a desync occurs, the robot's deviation is supposed to be *repaired* by the DT. To detect a desync, in each tick, the robot encodes its state into a verifier, which is either RS-ID's tag or SHA-256's partial digest. The verifier of size n_{ver} is transmitted to the DT.

In our scenario we neglect the transmission of the random position in RS-ID's position-tag pair, since we can assume a mechanism for common randomness (CR) generation between the parties as described in [12]–[14]. Common randomness also has been shown to increase the ID capacity of certain channels [15]. SHA-256 does not rely on randomness, thus only the verifier needs to be transmitted anyway.

The DT then uses the received verifier to identify whether its own state generated the same verifier, or not, thus performing a binary hypothesis test. If the verifiers match, the DT assumes no desync happened and does nothing, so the robot also assumes no desync happened and proceeds. If the verifiers do not match, the DT sends the state data of size n_{data} to the robot to repair its state. The data is assumed to be fully received before the next simulation tick. For error-free channels, the hypothesis test at the verifier cannot yield a false negative result. Thus, the DT will never falsely detect a desync and no unnecessary data will ever be transmitted to the robot. If the DT does not recognise a desync and erroneously verifies, no data is transmitted to the robot, and it remains in the desynced state. In the following tick, the states will remain out of sync since the parties assume different starting conditions. Since the probability of successive false positives deteriorates exponentially, only in very few occurrences the robot will remain out of sync for more than one tick. In summary, in each simulation tick, the verifier is transmitted. Only if the systems desynced (with p_{desync}) and the desync was not missed by the receiver (with p_{err} a function of the verifier size), the repair data is transmitted. Thus, per tick of our simplified model the expected optimal traffic and the expected traffic using ID are:

$$E(N_{\mathrm{opt}}) = p_{\mathrm{desync}} \cdot n_{\mathrm{data}} \tag{4}$$

$$E(N_{\mathrm{ID}}) = n_{\mathrm{ver}} + p_{\mathrm{desync}} \cdot (1 - \overline{p_{\mathrm{err}}(n_{\mathrm{ver}})}) \cdot n_{\mathrm{data}} \tag{5}$$

In real scenarios, p_{desync} is not fixed, but it depends on the environment and changes over time. Because real world states are not uniformly distributed, the ID decoder will see some ID pairs much more frequently than others. If p_{err} is unbound there are scenarios in which the most common pairs have a high error probability. By limiting p_{err} to λ_2 for all ID pairs, the number of errors will never grow to make the system inoperable, regardless of the underlying probability distribution of ID pairs, which is a strong argument in favor of using ID codes like RS-ID over using simple hashes. Finally, p_{desync} and p_{err} correlate in real scenarios, since the different states influence both the desync behavior and the error probability of the system.

To keep the latency as small as possible, the verification is only done at the DT. A simple extension of the scheme would introduce verification at both parties, reducing error rates at the cost of a small increase in computation, traffic, and delay.

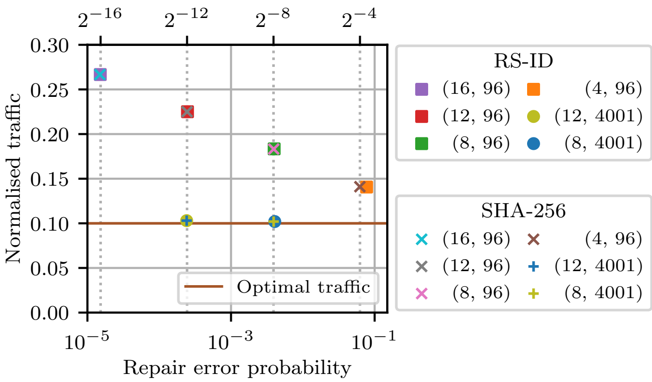

Fig. 2. Tuples of traffic and repair error probability for different configurations of $(n_{\text{ver}}, n_{\text{data}})$ at a desync probability of 0.1. The traffic is normalised by the traditional traffic. By increasing the size of the verifier, the repair error probability is reduced, but the traffic increases. The horizontal axis reports the repair error probability in base 10 and base 2.

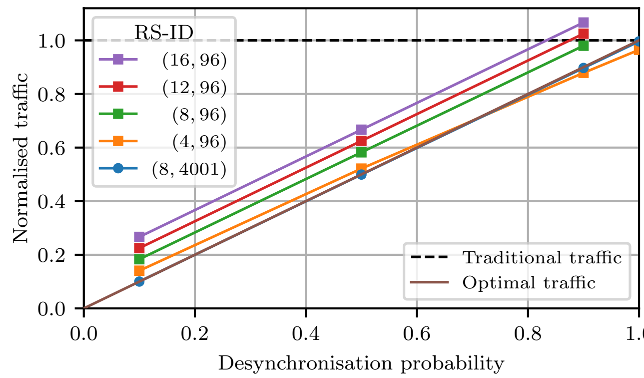

Fig. 3. Normalised traffic for different configurations of $(n_{\text{ver}}, n_{\text{data}})$ over p_{desync}. The traffic is normalised by the traditional traffic. SHA-256 and RS-ID show the same behaviour, so only RS-ID is shown.

B. Calculation of verifiers

In error correction applications, for which RS codes were designed, the full RS codeword has to be determined before its transmission. In RS-ID codes however, only one element of the codeword is required: the tag. Instead of calculating the complete RS codeword, it is sufficient to calculate the tag at the randomly chosen position. For this, we use discrete Fourier transforms (DFTs) for the RS encoding, which significantly reduces the complexity of tag computation [16], [17]. The finite field operations use the SageMath library, which allows for efficient finite field operations. Still, computation complexity of the RS-ID code remains very high, especially for high block lengths. For efficient SHA-256 computations we used the highly optimised Python library hashlib. To create a verifier of a certain bit length, only the first n_{ver} bits of the digest are transmitted. This allows for an efficient and simple implementation.

IV. Results

To verify the theory behind our approach, we carried out simulations using the $(n_{\text{ver}}, n_{\text{data}})$ RS-ID configurations shown in Table I. The SHA-256 approach uses the first n_{ver} bits of the digest and has no further parameters corresponding to the RS-ID code's. The double exponential behaviour of ID codes allows for a very large increase in the data size if the verifier size is increased. We chose the (4, 96) and (8, 4001) configurations to use almost all their representable IDs, while the other configurations reflect increasing the verifier size under constant data size. For comparability of different repair data sizes, we normalise the achieved traffic by the traditional traffic of always sending the repair data, but no verifiers.

Increasing the verifier size leads to a smaller average error probability at the cost of additional traffic as can be seen in Fig. 2. The traffic overhead over optimal traffic is given by $\frac{n_{\text{ver}}}{n_{\text{data}}}$. For a small data size, the overhead is thus quite large, while for the larger data size traffic exceeds optimal traffic by less than 1%. With one exception, RS-ID and SHA-256 show the same behavior across both traffic and average error probability. As the RS-ID code is generated in prime fields, we use the largest prime representable by n_{ver} bit for the RS base. For higher verifier sizes the deviation to SHA-256 is negligible, but at 4 bits this results in an average error of $\frac{1}{13}$ for RS-ID (1) and $\frac{1}{16}$ for SHA-256 (3). By increasing the data size to 4001 bit, the relative traffic introduced by the verifiers becomes negligible, while the average repair error probability is not effected at all. A larger verifier allows for fewer errors and more data simultaneously! As shown in Table I, even a small 16 bit verifier allows for states represented by millions of bits to be compared at very small error probabilities. Thus, ID codes are most effective for extremely large data sizes. To achieve small error probabilities you need to use larger verifier sizes, which become negligible in size if the scenario actually uses the full number of IDs representable by the verifier. For the given uniformly distributed set of IDs, which is also comparatively small in the double-exponential terms of ID, we did not find differences in performance between using RS-ID, which is optimal for ID, and the suboptimal SHA-256. In this scenario, they share the same average behavior.

The performance of our approach holds for other p_{desync} as well as seen in Fig. 3. For small data sizes our traffic can exceed traditional traffic under high p_{desync}, but for larger data sizes this only occurs in cases where the system desynchronises in almost every tick. For the smallest verifier size the traffic falls below optimal traffic for high p_{desync}, because p_{err} is quite high and the repair data is undesirably sent in fewer cases than necessary, as stated in equation 5. The (8, 4001) configurations approaches optimal traffic much closer than the other configurations, since the verifier is much smaller compared to the corresponding data, which can hold significantly more states. Still, the 8 bit configuration introduces enough errors for traffic to fall slightly below the optimal traffic for p_{desync} close to 1.

The effects of the bound error of RS-ID are not observed in our simulations. ID pairs with an average p_{err} contribute to far more errors than ID pairs with a high p_{err} close to λ_2.

In scenarios with some reliability constraints, the probability bound of ID codes is an advantage over the error-bound-less hashes at short block lengths.

The computation time of a verifier using the SHA-256 algorithm is at least an order of magnitude faster than computing a tag using RS-ID codes, as can be seen in Table II. While we have not optimised RS-ID, the difference is a major argument against using RS-ID codes in the described scenario. At short block lengths, the theoretically proven advantages of good ID codes do not translate into major practical benefits, and the additional computational costs involved do not pay off in relevant gains. The computation time outliers for the 4 bit verifiers are caused by the overhead of the other simulation parts not being outweighed by the verification process, which is the major computational expense at higher verifier sizes. Due to the great difference in computation complexity, we did not conduct any precise measurements.

TABLE II
AVERAGE COMPUTATION TIME PER VERIFIER

n_{ver}	4 bit	8 bit	12 bit	16 bit
RS-ID	36.3μs	10.7μs	11.5μs	11.2μs
SHA-256	1.35μs	0.28μs	0.25μs	0.25μs

V. Conclusion

While the extra computational cost of using RS-ID codes does not reflect in measurable advantages at the investigated configurations, SHA-256 offers a computationally cheap method to reduce traffic in non-mission-critical scenarios significantly. This reduces the bandwidth required in the wireless channels, potentially saving energy. The gains come at the cost of not synchronising the robot's and DT's state at all times. RS-ID codes are computationally more expensive than SHA-256, but they give higher reliability by bounding the repair error probability. We expect more complex scenarios to benefit from this bound. For ID codes to achieve their best behavior, the communicating parties need access to common randomness, which puts additional requirements on practical ID implementations. In future work, we will investigate k-ID [18] to extend our protocol to include more generalised scenarios and investigate computationally cheaper ID code constructions like [19].

Reducing traffic via identification offers several advantages. For the verifier transmission, one can use an energy-efficient low-rate channel. Only for the repairs, a high-rate channel is required. At low desync probabilities, several robots can communicate with the same MEC over a high-rate shared channel, further increasing the reduction in required bandwidth. Additionally, using verifiers reduces the delay of the DT learning whether the robot is in the desired state or not, since the DT only needs to receive a fraction of the full state's data to come to a verdict. The computational overhead of SHA-256 is below $1\mu s$ and, thus, no additional constraint to delay. Larger verifier sizes allow for an exponential reduction in repair error probability but put an additional traffic load on the network, which becomes only negligible at large state sizes. Identification codes work best for verifying extremely large states against each other.

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